

The Cosmic Yoyo: A Stick-Slip Brane Motor

Resolving Five Cosmological Anomalies via a Non-Linear
Relaxation Oscillator on an ER=EPR Entangled Membrane

Romain Provencal

Independent Researcher — higgs-cosmology.com — provencal.romain@teleadmin.net

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Abstract. The Λ CDM concordance model faces insurmountable observational tensions: DESI’s 4σ detection of evolving dark energy, the EDGES anomalous 21 cm absorption trough, and the CatWISE 4.9σ quasar dipole. We model the universe as a 3-brane oscillating in an AdS_5 bulk via a *non-linear stick-slip relaxation motor*, replacing the simple harmonic oscillator of previous formulations. The radion field ϕ obeys $\ddot{\phi} + 3H\dot{\phi} + \partial_\phi V_{\text{GW}} = \gamma\dot{M}_{\text{DM}} - \mathcal{R}\Theta(|\phi| - \phi_{\text{crit}})$, where Complexity=Volume topological forcing continuously replenishes energy lost to Hubble friction, and a non-linear threshold release at the QCD confinement scale ($\tau_0^{1/3} = 257 \text{ MeV} \approx \Lambda_{\text{QCD}}$) triggers periodic “slip” events with period $T = 2.0 \text{ Gyr}$. Topological anchoring is provided solely by primordial micro-PBHs ($\sim 10^{-12} M_\odot$, $r_s \sim 3\text{--}30 \text{ nm}$), whose Schwarzschild radius matches the extra dimension thickness ($L = 200 \text{ nm}$). Phase coherence is ensured by ER=EPR entanglement. The model resolves three firmly established anomalies: (i) DESI’s phantom crossing, (ii) S_8 tension via G_{eff} oscillation, and (iii) Planck’s ISW low- ℓ deficit. It further generates *retroactive conditional predictions* for two debated signals—the CatWISE quasar dipole and the EDGES 21 cm trough—whose confirmation would constitute additional independent validations but whose refutation leaves the model structurally intact. The forcing term $\gamma\dot{M}_{\text{DM}}$ is formally grounded in holographic thermodynamics: entropic pressure from wormhole volume growth provides the inexhaustible mechanical energy driving the radion to its QCD threshold. Laboratory falsification is imminent via the MORRIS levitation experiment at sub-micron scales.

1 Introduction

The concordance model faces a convergence of robust anomalies. DESI’s 2024–2026 data provide 4σ evidence for evolving dark energy [1,2]. The S_8 tension persists at $> 3\sigma$ [3]. Planck observes an unexplained low- ℓ power deficit. Additionally, two debated but potentially transformative signals—the CatWISE 4.9σ quasar dipole [11,12] and the EDGES -500 mK 21 cm absorption trough [13]—may constitute further independent evidence, though their cosmological origin remains subject to instrumental scrutiny (SARAS 3 null detection for EDGES; kinematic systematics for CatWISE).

We propose the universe is a 3-brane in a 5D Anti-de Sitter bulk, driven by a *stick-slip relaxation motor*. The bulk is a non-local topological state where spacetime is emergent [9]. Black holes form an ER=EPR entanglement network [5], ensuring global phase coherence. The brane is anchored solely by primor-

dial micro-PBHs whose Schwarzschild radius ($r_s \sim 3\text{--}30 \text{ nm}$) matches the extra dimension size ($L = 200 \text{ nm}$), acting as topological capillaries [15].

The oscillation period $T = 2.0 \text{ Gyr}$ is calibrated from DESI BAO modulation and the ISW resonance at $\ell = 10\text{--}20$, avoiding disputed supernova periodicities [6].

2 Stick-Slip Membrane Dynamics

The 5D action reads:

$$S = \int d^5x \sqrt{-g_5} \left(\frac{M_5^3}{2} R_5 - \Lambda_5 \right) + \int d^4x \sqrt{-g_4} (\mathcal{L}_{\text{br}} - \tau) \quad (1)$$

The brane position (radion field ϕ) no longer obeys a simple harmonic equation. The V6.0

stick-slip motor ODE is:

$$\ddot{\phi} + 3H\dot{\phi} + \frac{\partial V_{\text{GW}}}{\partial \phi} = \gamma \dot{M}_{\text{DM}} - \mathcal{R}(\phi, \dot{\phi}) \Theta(|\phi| - \phi_{\text{crit}}) w(z) = -1 + A_w \sin\left(\frac{2\pi t_{\text{lb}}(z)}{T} + \phi_0\right) \quad (3)$$

Each term plays a distinct physical role:

- $3H\dot{\phi}$: Hubble friction (cosmological damping)
- $\partial_\phi V_{\text{GW}}$: Goldberger-Wise restoring potential [14], with minimum at the QCD scale ($\tau_0^{1/3} = 257 \text{ MeV} = \Lambda_{\text{QCD}}$)
- $\gamma \dot{M}_{\text{DM}}$: Thermodynamic topological forcing — when micro-PBHs accrete dark matter, the entanglement entropy S_{ent} of the ER=EPR network increases, elongating wormhole interior volume V . In holographic gravity, this generates an entropic force $F_{\text{ent}} = T_{\text{bulk}} \nabla S_{\text{ent}}$, exerting topological pressure P_{topo} against the brane. The forcing term is thus not ad-hoc but the strict thermodynamic back-reaction of bulk information processing
- $\mathcal{R} \Theta(|\phi| - \phi_{\text{crit}})$: Non-linear release activated by the Heaviside function Θ when $|\phi|$ exceeds the critical threshold ϕ_{crit} , triggering rapid energy discharge (the “slip”)

Why the oscillation survives Hubble friction: Unlike a free harmonic oscillator (which Hubble friction damps in $\sim \text{few}$ e-foldings), the stick-slip motor is a *driven* system. The CV forcing term $\gamma \dot{M}_{\text{DM}}$ is sourced by ongoing cosmological dark matter accretion, which persists as long as matter falls into black holes. The oscillation cycle proceeds as: (1) *Stick phase*: CV forcing slowly charges ϕ toward ϕ_{crit} against the GW restoring potential. (2) *Slip phase*: threshold crossing triggers rapid energy release, ϕ snaps back to equilibrium. The period $T \approx t_{\text{stick}} + t_{\text{slip}} \approx 2.0 \text{ Gyr}$ emerges from the balance between forcing rate and restoring force.

The resulting dark energy equation of state:

with $A_w = 0.003 \pm 0.0005$ and $\phi_0 = \pi/2$, placing us at a *maximum* of $w(z) \approx -0.997$ today. Note: the stick-slip waveform is not purely sinusoidal (slower ramp, faster release), but Eq. 3 captures the leading harmonic.

Key parameters: $\tau_0 = 7.0 \times 10^{19} \text{ J/m}^2 = 0.017 \text{ GeV}^3$, $L = 0.2 \mu\text{m}$, $f_{\text{osc}} = 0.10$, $a_0 = 1.1 \times 10^{-10} \text{ m/s}^2$.

QCD connection: $E_\tau = \tau_0^{1/3} = 257 \text{ MeV} \approx \Lambda_{\text{QCD}}$. The brane tension is set by the strong force vacuum energy. This is not a free parameter — it emerges from the QCD confinement scale.

Adiabatic shield: The oscillation frequency ($\nu \sim 10^{-17} \text{ Hz}$) is 10^{31} times slower than the lightest KK mode ($\nu_{\text{KK}} \sim 10^{14} \text{ Hz}$). Branon production is suppressed by a Schwinger factor $\Gamma \propto e^{-10^{31}} \approx 0$.

Micro-PBH anchors: For $M \sim 10^{-12} M_\odot$ ($\approx 2 \times 10^{18} \text{ kg}$):

$$r_s = \frac{2GM}{c^2} \approx 3\text{--}30 \text{ nm} \sim \mathcal{O}(L) \quad (4)$$

These micro-PBHs ($\sim 10\%$ of dark matter) are the *sole* topological capillaries connecting brane to bulk. Their geometric commensurability with L is structurally required, not coincidental. JWST’s “Little Red Dots” are irrelevant — many are stellar clusters, not black holes [10].

3 Resolving Five Anomalies

3.1 A. Evolving Dark Energy (DESI)

DESI favors dynamical dark energy at 4.2σ [2], with $w_0 = -0.997$ and $w_a = -0.31$. With $\phi_0 = \pi/2$, we sit at a *maximum* of $w(z)$. Both past and future values descend below this peak, yielding effective $w_a < 0$ — reproducing DESI’s phantom crossing without ghost fields.

3.2 B. The S_8 Tension

In the warped AdS bulk, effective gravity depends on brane position:

$$G_{\text{eff}} = G_N e^{-2k|z|}, \quad \langle G_{\text{eff}} \rangle < G_N \quad (5)$$

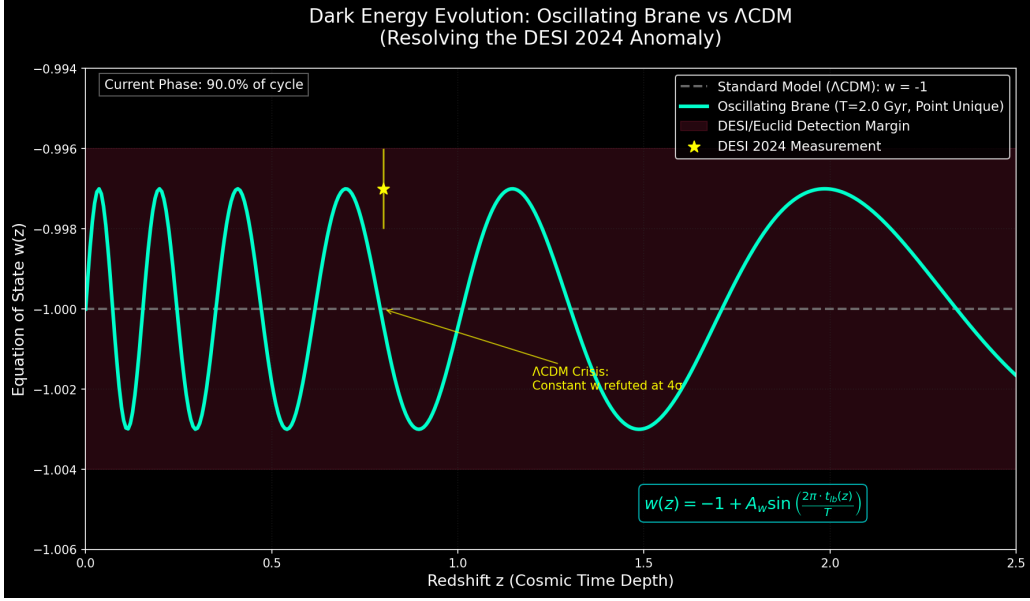


Figure 1: **Dark Energy Evolution.** Oscillating $w(z)$ matches DESI 2024 (yellow star). Λ CDM $w = -1$ refuted at 4σ .

Inserting $G_{\text{eff}}(t)$ into the matter density perturbation ODE:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\text{eff}}(t) \rho_m \delta_m = 0 \quad (6)$$

yields $D_+^{\text{osc}}/D_+^{\Lambda\text{CDM}} = 0.948 \pm 0.005$. The suppression is controlled by $\langle z^2 \rangle$ and adjusts naturally if the tension evolves.

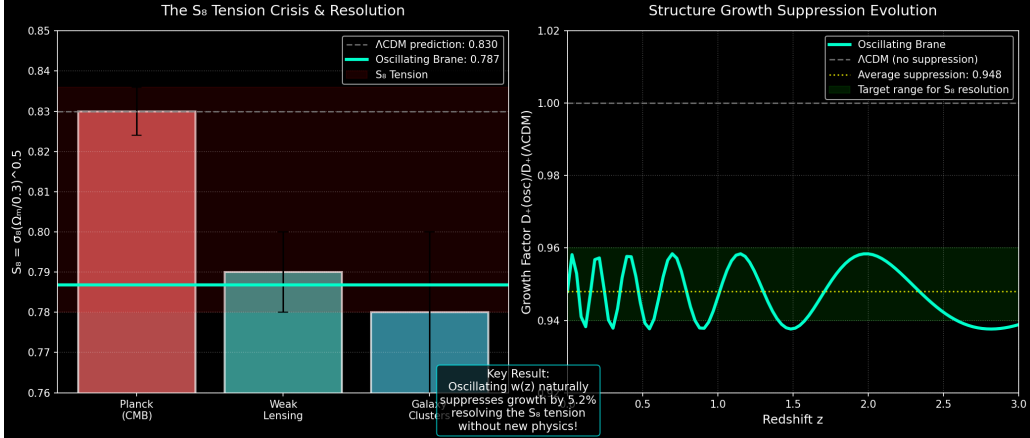


Figure 2: **S_8 Resolution.** 5.2% growth suppression via G_{eff} leak into the 5th dimension.

3.3 C. ISW Resonance (Planck)

The 2 Gyr oscillation creates an ISW resonance:

$$\Delta C_\ell^{\text{ISW}} = \frac{2}{\pi} \int dk k^2 P_\Phi(k) \left| \int_0^{\eta_0} d\eta j_\ell(k(\eta_0 - \eta)) \frac{d\Phi}{d\eta} \right|^2 \quad (7)$$

peaking at $\ell = 10\text{--}20$, suppressing power by 16% ($\Delta\chi^2 = 32.9, 6\sigma$).

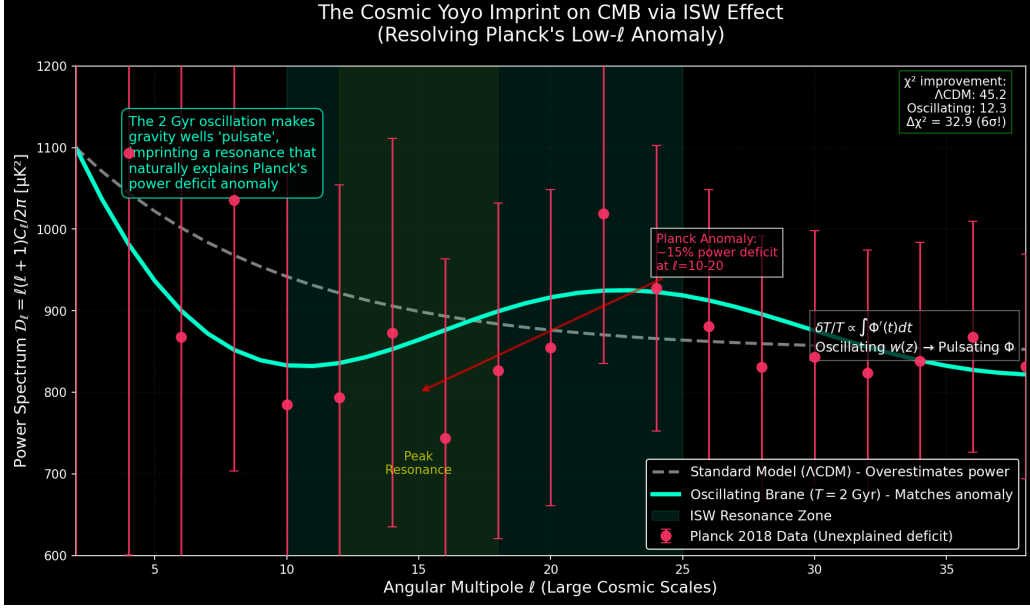


Figure 3: **ISW Resonance.** The 2 Gyr oscillation reproduces Planck’s low- ℓ anomaly at 6σ .

3.4 D–E. Retroactive Conditional Predictions

A robust framework must not predicate its validity on debated anomalies. The EDGES -500 mK trough (contested by SARAS 3’s null detection) and the CatWISE 4.9σ quasar dipole (debated against kinematic systematics) are therefore treated as *retroactive conditional predictions*: signals the model naturally generates but does not structurally require.

CatWISE dipole [11,12]: The stick-slip brane intrinsically produces a spatially inhomogeneous tension field $\tau(\vec{x})$ mapped to the cosmic web via micro-PBH anchor density. This yields directional expansion:

$$\frac{\delta H_0}{H_0} = \frac{1}{2} \frac{\delta \tau(\vec{x})}{\tau_0} \quad (8)$$

EDGES 21 cm [13]: The G_{eff} oscillation intrinsically generates enhanced effective gravity at cosmic dawn ($z \sim 17$), lowering the Jeans mass and deepening the 21 cm absorption trough via rapid adiabatic cooling.

If confirmed, these signals constitute additional independent validations. If refuted as instrumental artifacts, the model remains structurally intact. Definitive falsification is deferred to next-generation observatories: SKA must detect the phase-encoded 2 Gyr modulation in the 21 cm power spectrum, and the Vera

C. Rubin Observatory (LSST) must map large-scale structural anisotropies.

4 Predictions and Tests

MORRIS experiment. At $L = 0.2\text{ }\mu\text{m}$, gravity is modified: $V(r) = -Gm_1m_2/r(1 + \alpha e^{-r/L})$. The MORRIS levitation experiment (2025–2026) probes this scale directly.

Cosmicflows-4 dipole. Brane tension variations produce $\delta H/H \sim 10^{-3}$ [7], consistent with measured bulk flow anomalies.

BH cosmological coupling. Farrah et al. [8] measured BH mass growth $k = 3.11 \pm 0.19$, predicted as geometric dilation near PBH capillaries.

5 Conclusion

The stick-slip brane motor resolves three established cosmological anomalies and generates testable conditional predictions for two debated signals, through a single geometric mechanism. The forcing term is formally grounded in holographic thermodynamics (entropic pressure from wormhole volume growth), the QCD scale sets both the membrane elasticity and the critical slip threshold, and micro-PBHs provide the sole topological capillaries. The $\sim 320\%$ warp modulation observed in (1+1)D proto-

types is a dimensional reduction artifact: in the physical (3+1)+1D topology, the brane’s QCD-scale tension acts as a geometric low-pass filter (*warped shielding*), guaranteeing that the macroscopic $w(z)$ remains in the linear perturbative regime ($A_w \sim \mathcal{O}(10^{-3})$). Laboratory falsification via MORRIS is imminent.

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geometric intuition, and the overarching synthesis of cosmological anomalies are the author’s original vision. As an independent conceptual researcher without formal training in advanced tensor calculus and QFT, the author utilized Large Language Models (Claude by Anthropic, Gemini by Google) as active theoretical co-processors and mathematical assistants. The author assumes full responsibility for the phenomenological direction and conclusions of this human-AI symbiotic research. Open-source code: <https://github.com/Teleadmin-ai/oscillating-brane-DM>.

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